

Ricardo Valero

Senior Software Engineer

I build typed, declarative systems: effects that declare their requirements and failures, schemas at every boundary, builds that reproduce anywhere. I work spec-first with coding agents, writing the specification and letting the implementation follow, and I keep the tooling fast and light.

ricardo-valero ricardo-valero-de-la-rosa

WORK EXPERIENCE

Manuable, S.A.P.I. de C.V.

Senior Software Engineer

04.2025 → Present

Multi-provider Shipping Platform

I found the provider logic duplicated across three places, with business rules tangled into application code, and led the refactor that pulled them apart. Adding a carrier or fixing a broken one went from about two weeks to about a day. I review the work of the three engineers building in this codebase.

- Provider integrations (FedEx, DHL, Estafeta): rate pricing and margins, label recovery flows, and error surfacing
- UI components across the product, and N+1 query fixes on the slowest paths

Ruby Rails PostgreSQL JavaScript

Integration Hub: Provider Shipment Service

Sole author. I shaped the public contract on Stripe's event envelope and idempotency keys, so consumer apps integrate once regardless of which carrier is behind the call, and made the pure/IO split structural: carrier modules declare no HTTP library, so the compiler refuses a network call from inside one.

- Standalone service executing ship-quote, rate, and label-generation calls for multiple consumer applications, across eight carriers
- Stateless on tenancy: provider credentials exchanged for encrypted session bearers, never persisted at rest
- 73 test files, with CI gating formatting, Credo, Dialyzer, and the suite on every push
- Delivered spec-first: 47 specified changes in the first seven weeks

Elixir Phoenix PostgreSQL Oban OpenAPI Nix

Data Hub: Centralized Ingestion and Transformation

Earlier engineers had written the fragmented ETL off as debt to live with. I designed the replacement and ran the migration alone, and the engineers who joined afterwards were shipping against it inside a week.

- Ingestion pipelines landing source systems into BigQuery with a SQLMesh transformation layer, as a data-stack monorepo
- Consolidated earlier fragmented ETL services (Airbyte, ad-hoc GCP jobs) into a single platform

Python dlt SQLMesh BigQuery GCP Metabase

Invoice Data Ingestion: Spreadsheets to BigQuery

I took the input nobody wants to own: invoices arriving as inconsistent multi-sheet workbooks, no two quite alike, and turned them into typed rows the downstream models could depend on.

- Transformed raw Excel, CSV, and spreadsheet invoice data into structured BigQuery ingestions with multi-sheet imports

Elixir Phoenix libSQL BigQuery

Infrastructure Tooling

I made a flake.nix a requirement on every project in the company and moved our CI on both Bitbucket and GitHub to build through Nix, so a build behaves the same on a laptop as it does in CI.

- Built a Nix-managed CLI deploying a self-hosted PaaS (Dokploy) and compose services across AWS and GCP
- Fleet management for company services: machines, DNS, secrets, and deploys, one VM per app

Nix NixOS Terraform AWS GCP Docker

Hitower Technohub, S. de R.L. de C.V.

Senior Software Engineer

09.2024 → 02.2025

SEO and Content Optimization Web Service (Write2Rank)

I owned the analysis path end to end, from the scoring APIs through the interface that had to explain a ranking suggestion well enough for a writer to act on it.

- Full-stack web application for content analysis and ranking suggestions
- Backend APIs and frontend interfaces designed to scale with content volume

PHP

Laravel

PostgreSQL

TypeScript

Node.js

Next.js

CI/CD Pipeline Optimizations

I brought reproducible builds into a pipeline that did not have them, so a green build stopped depending on which machine produced it, and cut build times through caching and container changes.

- Experiments to improve build times and deployment processes
- Reproducible builds, and caching and container usage tuned for faster CI/CD

AWS

GitLab CI

Docker

Nix

Diseño Control y Sistemas a la Medida, S.A. de C.V. (aka Industrial Code)

Senior Software Engineer

04.2021 → 05.2023

Assembly Line Real-time Production Monitoring Web Application for Polaris Inc.

I built the view of the whole line: many live signal sources and APIs converging into one screen an operator could read at a glance while the line was running.

- Led design and architecture of UI/UX, tech stack, database, and APIs

TypeScript

Node.js

Fastify

tRPC

DrizzleORM

PostgreSQL

React

Production Reporting Platform (SCADA/MES) for Polaris Inc.

The reporting charts degraded as production history piled up. I redesigned the schema and the indexes behind them so the reports stayed usable as the data grew, normalized what the old system had flattened, and carried the migration out against live production data.

- Reporting and charting over accumulated assembly-line production data
- SQL Server schema, index design, and normalization; migration performed on live data
- Migrated the application from .NET Framework and EF6 to .NET Core and EF Core

C#

.NET Core

.NET Framework

ASP.NET MVC

Razor

EF Core

EntityFramework

MSSQL

Vehicle Registration Hybrid Web and Mobile Application (Registegic)

First hybrid web and mobile build in the company and my first mobile target, and I led three junior engineers through it.

- Hybrid web and mobile delivery from a single codebase

AWS

TypeScript

Node.js

Fastify

tRPC

Prisma

PostgreSQL

React

Warehouse Repackaging Control Web Application for Polaris Inc.

I drove the serial hardware straight from Chrome over Web Serial, with readings streaming live into React, so the floor needed nothing installed on the machine to scan and repack.

- Scan, repack, and confirmation flow tied to floor hardware

TypeScript

Node.js

Fastify

GraphQL

Prisma

PostgreSQL

React

WebSerial

Hardware Integration for IoT

I designed the adapter layer that let one API speak to serial devices, printers, tools, and sensors that each had their own protocol and none of which agreed on anything.

- Designed tech stack and APIs for serial adapters, printers, tools, and IoT sensors

TypeScript

Node.js

SQLite

Tecnología en Sistemas de Integración S.A. de C.V. (TSI)

Junior Software Engineer

01.2019 → 02.2021

Assembly Line Control Applications for Polaris Inc.

My first production systems: the code that pulled data off the tools on the line and made it something the plant could act on.

- Custom SCADA/MES, hardware integration, data collection, database design, and APIs

VBScript

Indusoft

REST

OpenProtocol

ODBC

Work Shift Management Web Application for Polaris Inc.

Where I learned .NET and SQL Server properly, by modelling crews and shifts that everything downstream then had to agree with.

- Full-stack application, database design, and APIs

C#

.NET

REST

EntityFramework

MSSQL

React

Assembly Line Reporting Web Application for Polaris Inc.

My first reporting work: turning raw line output and downtime into the numbers supervisors actually ran the shift on.

- Shift and line output reporting over collected production data

C#

.NET

REST

EntityFramework

MSSQL

Razor

Internship in Automation Engineering

08.2018 → 12.2018

Ladder-logic PLC Program Development

Where I started: writing control logic that moved physical machinery, on hardware where a mistake stops a plant rather than a process.

- Several ladder-logic PLC programs for industrial clients, on Honeywell and Rockwell controllers

Ladder Logic

Honeywell PLC

Rockwell PLC

Institut für Mikrostrukturtechnik (IMT), Karlsruher Institut für Technologie (KIT)

Internship in Advanced Materials and Optical Spectroscopy

03.2017 → 07.2017

Automated Dip-coating Control for SURMOF Deposition

My first automation of an experiment: the timing program that made a deposition run repeatable, so results could be compared between runs rather than between operators.

- MATLAB timing control for a dip-coating machine in solar-cell research: immersion, hold, withdrawal, and transfer between liquid baths
- Optimized thin-film deposition for large-area SURMOF layers

MATLAB

SIDE PROJECTS

Deplugger: Type-safe Workflow Platform

- Designed an op/workflow DSL where workflows are data: every op declares input/output schemas, so composed workflows are validated end-to-end at every step
- Built a dual-target compiler: the same workflow runs as an ephemeral effect or compiles to a durable, replay-safe workflow with activity semantics
- Streaming RPC surfaces live run telemetry to the UI; credentials encrypted at rest with schema-derived secret redaction

TypeScript

Effect

Bun

SolidJS

SQLite

Nix

Cloudflare

Raggio: Schema Library for Elixir

- Port of Effect-TS Schema to Elixir: composable, bidirectional schemas for decoding, encoding, and validation with typed errors
- Adopted in production as the domain schema layer of a company service, replacing a hand-rolled DSL

Elixir

Open-source Libraries for Roc and Effect

- Ecosystem libraries for the Roc language: parser combinators, JSON, Unicode, ANSI, GraphQL, PostgreSQL, common data structures, and an experimental VS Code extension
- A browser platform for framebuffer apps targeting wasm32, with a Zig host and WebGPU
- NES, Game Boy, and PlayStation 1 emulators written in pure Roc, as a test of the language under a demanding workload; the R3000A core passes all 55,000 SingleStepTests cases
- Earlier work across the Effect TypeScript ecosystem: runtime and data-type experiments, a PostgreSQL layer, and an ESLint rule set

Roc TypeScript Effect Zig WASM WebGPU Nix

EDUCATION

Universidad Autónoma de Nuevo León (UANL), Facultad de Ingeniería Mecánica y Eléctrica (FIME)

08.2013 → 12.2018

Bachelor in Mechanical and Electrical Engineering
Specialization in Thermodynamics and Refrigeration

Karlsruher Institut für Technologie (KIT)

08.2016 → 07.2017

Academic Exchange in Mechanical Engineering
Scholarship from the German Academic Exchange Service (DAAD)

SKILLS

Languages	TypeScript, C#, Python, Ruby, Elixir, PHP, SQL
AI-Assisted Engineering	Spec-Driven Development, OpenSpec, Claude Code, opencode, Agent skill authoring, Structured prompts and specifications, Review of AI-generated code
Backend	.NET, ASP.NET, EntityFramework, Node.js, Bun, Deno, Express, Fastify, tRPC, Effect, DrizzleORM, Prisma, Rails, Phoenix, Laravel
Frontend	React, Next.js, SolidJS, Svelte, WebAPIs, WASM
Databases	MSSQL, PostgreSQL, MariaDB, SQLite
Testing	ExUnit, Playwright, happy-dom, Credo, Dialyzer
DevOps	AWS, GCP, Docker, Nix, Terraform, GitHub Actions, GitLab CI, Bitbucket Pipelines
Data	dlt, SQLMesh, BigQuery, Metabase
Protocols	REST, GraphQL, gRPC, HTTP, WebSockets, SSH, OpenProtocol, ODBC
Daily Drivers	macOS, Ghostty, Zed, Nix, direnv

LANGUAGES

Spanish	native
English	advanced
German	basic